

```
In [1]: import torch
torch.cuda.empty_cache()
torch.cuda.ipc_collect()
```

```
In [2]: import os
os.environ["PYTORCH_CUDA_ALLOC_CONF"] = "expandable_segments:True"
```

```
In [ ]: !pip install torch torchvision torchaudio --index-url https://download.pytorch.o
```

```
In [ ]: !pip install transformers datasets peft bitsandbytes accelerate scikit-learn -U
```

```
In [1]: import torch

print("CUDA available:", torch.cuda.is_available())

if torch.cuda.is_available():
    print("GPU name:", torch.cuda.get_device_name(0))
```

CUDA available: True

GPU name: NVIDIA RTX A6000

```
In [2]: # =====
# IMPORTS
# =====

import pandas as pd
import numpy as np
import gc
import torch
import torch.nn as nn

from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score,
    f1_score,
    classification_report,
    confusion_matrix
)

from sklearn.utils.class_weight import compute_class_weight

from datasets import Dataset

from transformers import (
    AutoTokenizer,
    AutoModelForCausalLM,
    TrainingArguments,
    Trainer, BitsAndBytesConfig
)

from peft import prepare_model_for_kbit_training, LoraConfig, get_peft_model
```

```
/opt/conda/lib/python3.10/site-packages/tqdm/auto.py:21: TqdmWarning: IPProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
from .autonotebook import tqdm as notebook_tqdm
```

```

In [3]: # =====
# LOAD BALANCED DATASETS
# =====

course_balanced = pd.read_csv(
    "course_balanced-new.csv"
)

teacher_balanced = pd.read_csv(
    "teacher_balanced-new.csv"
)

university_balanced = pd.read_csv(
    "university_balanced-new.csv"
)

# =====
# LOAD SHARED FIXED TEST SETS
# =====

teacher_test = pd.read_csv(
    "teacher_test.csv"
)

course_test = pd.read_csv(
    "course_test.csv"
)

university_test = pd.read_csv(
    "university_test.csv"
)

```

```

In [4]: train_df = pd.concat(
    [course_balanced, teacher_balanced]
).reset_index(drop=True)

test_df = university_test.reset_index(drop=True)

print("Train:", train_df.shape)
print("Test:", test_df.shape)

```

Train: (8436, 5)

Test: (844, 7)

```

In [5]: # =====
# TRAIN / VALIDATION SPLIT
# =====

train_df, val_df = train_test_split(
    train_df,
    test_size=0.1111,
    stratify=train_df["sentiment"],
    random_state=42
)

```

```

train_df = train_df.reset_index(drop=True)
val_df = val_df.reset_index(drop=True)

# FIXED SHARED TEST SET
test_df = test_df.reset_index(drop=True)

print("Train:", train_df.shape)
print("Validation:", val_df.shape)
print("Test:", test_df.shape)

```

Train: (7498, 5)
 Validation: (938, 5)
 Test: (844, 7)

```

In [6]: # =====
# LOGIN
# =====
from huggingface_hub import login
hf_token = input("Enter your Hugging Face token: ")
login(hf_token)

```

```

In [7]: # =====
# LABEL MAPPING
# =====

label2id = {
    "negative": 0,
    "neutral": 1,
    "positive": 2
}

id2label = {
    0: "negative",
    1: "neutral",
    2: "positive"
}

train_df["label"] = train_df["sentiment"].map(label2id)
val_df["label"] = val_df["sentiment"].map(label2id)
test_df["label"] = test_df["sentiment"].map(label2id)

```

```

In [8]: # =====
# CLASS WEIGHTS
# =====

classes = np.array([0, 1, 2])

class_weights = compute_class_weight(

    class_weight="balanced",

    classes=classes,

    y=train_df["label"].values
)

class_weights = torch.tensor(
    class_weights,
    dtype=torch.float
)

```

```
)

print("\nClass Weights:")
print(class_weights)
```

Class Weights:
tensor([0.9721, 2.4265, 0.6413])

```
In [9]: # =====
# FORMAT INSTRUCTION
# =====

def format_instruction(row):

    return f"""<s>[INST] You are an expert sentiment classifier.

Classify the sentiment into:
positive, negative, or neutral.

Only output ONE word.

Review:
{row['review']}

Aspect:
{row['aspect']}

Answer:
[/INST] {row['sentiment']}"""
```

```
In [10]: # =====
# CREATE TEXT
# =====

train_df["text"] = train_df.apply(
    format_instruction,
    axis=1
)

val_df["text"] = val_df.apply(
    format_instruction,
    axis=1
)

test_df["text"] = test_df.apply(
    format_instruction,
    axis=1
)
```

```
In [11]: # =====
# TOKENIZER
# =====

model_name = "meta-llama/Meta-Llama-3-8B-Instruct"

tokenizer = AutoTokenizer.from_pretrained(
    model_name,
    token=True
)
```

```
tokenizer.pad_token = tokenizer.eos_token
```

```
In [12]: # =====
# TOKENIZATION
# =====

def tokenize_function(example):

    full_text = example["text"]

    prompt, answer = full_text.split("[/INST]")

    prompt = prompt + "[/INST]"

    answer = answer.strip()

    prompt_ids = tokenizer(
        prompt,
        truncation=True,
        max_length=128
    )["input_ids"]

    answer_ids = tokenizer(
        answer,
        truncation=True,
        max_length=10
    )["input_ids"]

    input_ids = prompt_ids + answer_ids

    labels = (
        [-100] * len(prompt_ids)
        + answer_ids
    )

    max_len = 128

    input_ids = input_ids[:max_len]

    labels = labels[:max_len]

    padding_length = max_len - len(input_ids)

    input_ids += (
        [tokenizer.pad_token_id]
        * padding_length
    )

    labels += (
        [-100]
        * padding_length
    )

    attention_mask = [

        1 if token != tokenizer.pad_token_id else 0

        for token in input_ids
    ]
```

```

return {

    "input_ids": input_ids,

    "attention_mask": attention_mask,

    "labels": labels,

    "class_label": example["label"]

}

```

```

In [13]: # =====
# DATASETS
# =====

train_dataset = Dataset.from_pandas(
    train_df[["text", "label"]]
)

val_dataset = Dataset.from_pandas(
    val_df[["text", "label"]]
)

test_dataset = Dataset.from_pandas(
    test_df[["text", "label"]]
)

train_dataset = train_dataset.map(
    tokenize_function,
    remove_columns=["text", "label"]
)

val_dataset = val_dataset.map(
    tokenize_function,
    remove_columns=["text", "label"]
)

test_dataset = test_dataset.map(
    tokenize_function,
    remove_columns=["text", "label"]
)

train_dataset.set_format(type="torch")

val_dataset.set_format(type="torch")

test_dataset.set_format(type="torch")

```

```

Map: 100%|██████████| 7498/7498 [00:06<00:00, 1227.72 examples/s]
Map: 100%|██████████| 938/938 [00:00<00:00, 1182.06 examples/s]
Map: 100%|██████████| 844/844 [00:00<00:00, 1141.00 examples/s]

```

```

In [14]: # =====
# CUSTOM TRAINER
# =====

class WeightedTrainer(Trainer):

```

```
def compute_loss(
    self,
    model,
    inputs,
    return_outputs=False,
    **kwargs
):
    labels = inputs.pop("labels")
    class_label = inputs.pop("class_label")
    outputs = model(**inputs)
    logits = outputs.logits
    shift_logits = logits[:, :-1, :].contiguous()
    shift_labels = labels[:, 1:].contiguous()
    loss_fct = nn.CrossEntropyLoss(
        reduction="none"
    )
    loss = loss_fct(
        shift_logits.view(
            -1,
            shift_logits.size(-1)
        ),
        shift_labels.view(-1)
    )
    loss = loss.view(
        shift_labels.size()
    )
    mask = shift_labels != -100
    loss = loss * mask
    sequence_loss = (
        loss.sum(dim=1)
        / mask.sum(dim=1)
    )
    weights = class_weights.to(
        class_label.device
    )[class_label]
    sequence_loss = (
```

```

        sequence_loss * weights
    )

    loss = sequence_loss.mean()

    return (
        (loss, outputs)
        if return_outputs
        else loss
    )

```

```

In [15]: # =====
# EXPERIMENT FUNCTION
# =====

def run_experiment(
    experiment_name,

    learning_rate=1e-5,

    epochs=5
):

    gc.collect()

    torch.cuda.empty_cache()

    # =====
    # QUANTIZATION
    # =====

    bnb_config = BitsAndBytesConfig(

        load_in_4bit=True,

        bnb_4bit_compute_dtype=torch.float16,

        bnb_4bit_use_double_quant=True,

        bnb_4bit_quant_type="nf4"
    )

    # =====
    # LOAD MODEL
    # =====

    gc.collect()
    torch.cuda.empty_cache()
    torch.cuda.ipc_collect()

    model = AutoModelForCausalLM.from_pretrained(
        model_name,
        quantization_config=bnb_config,
        device_map="auto",
        torch_dtype=torch.float16,
        low_cpu_mem_usage=True
    )

```

```

# =====
# PREPARE MODEL
# =====

model = prepare_model_for_kbit_training(
    model
)

# =====
# LORA CONFIG
# =====

lora_config = LoraConfig(

    r=8,

    lora_alpha=16,

    target_modules=[
        "q_proj",
        "v_proj"
    ],

    lora_dropout=0.05,

    bias="none",

    task_type="CAUSAL_LM"
)

# =====
# ADD LORA
# =====

model = get_peft_model(
    model,
    lora_config
)

model.config.use_cache = False

model.gradient_checkpointing_enable()

model.train()

# =====
# TRAINING ARGS
# =====

training_args = TrainingArguments(

    output_dir=f"./{experiment_name}",

    per_device_train_batch_size=1,

```

```

        gradient_accumulation_steps=8,

        num_train_epochs=epochs,

        learning_rate=learning_rate,

        logging_steps=20,

        save_strategy="epoch",

        save_total_limit=1,

        fp16=True,

        gradient_checkpointing=True,

        report_to="none",

        remove_unused_columns=False,

        optim="paged_adamw_8bit",

        max_grad_norm=1.0
    )

# =====
# TRAINER
# =====

trainer = WeightedTrainer(

    model=model,

    args=training_args,

    train_dataset=train_dataset,

    eval_dataset=val_dataset
)

# =====
# TRAIN
# =====

trainer.train()

# =====
# INFERENCE FUNCTION
# =====

def predict(df):

    model.eval()

    preds = []

    labels = []

```

```

device = next(model.parameters()).device

for i in range(len(df)):

    review = df.iloc[i]["review"]

    aspect = df.iloc[i]["aspect"]

    true_label = df.iloc[i]["sentiment"]

    prompt = f"""<s>[INST] You are an expert sentiment classifier.

Classify the sentiment into:
positive, negative, or neutral.

Only output ONE word.

Review:
{review}

Aspect:
{aspect}

Answer:
[/INST]"""

    inputs = tokenizer(

        prompt,

        return_tensors="pt",

        truncation=True,

        max_length=128

    ).to(device)

    with torch.no_grad():

        output = model.generate(

            **inputs,

            max_new_tokens=3,

            do_sample=False,

            temperature=0.0,

            pad_token_id=tokenizer.eos_token_id

        )

    decoded = tokenizer.decode(

        output[0],

        skip_special_tokens=True

    )

```

```

        response = decoded.split(
            "[/INST]"
        )[-1].strip().lower()

        if "positive" in response:

            pred = "positive"

        elif "negative" in response:

            pred = "negative"

        elif "neutral" in response:

            pred = "neutral"

        else:

            pred = "neutral"

        preds.append(pred)

        labels.append(true_label)

    return preds, labels

# =====
# EVALUATION
# =====

print("\n===== CROSS DOMAIN RESULTS =====")

preds_cross, labels_cross = predict(test_df)

accuracy = accuracy_score(
    labels_cross,
    preds_cross
)

macro_f1 = f1_score(
    labels_cross,
    preds_cross,
    average="macro"
)

weighted_f1 = f1_score(
    labels_cross,
    preds_cross,
    average="weighted"
)

print("Accuracy:", accuracy)

print("Macro F1:", macro_f1)

print("Weighted F1:", weighted_f1)

print("\nClassification Report:\n")

```

```
print(
    classification_report(
        labels_cross,
        preds_cross
    )
)

# =====
# SAVE PREDICTIONS
# =====

final_predictions_df = test_df.copy()

final_predictions_df[
    "true_sentiment"
] = labels_cross

final_predictions_df[
    "predicted_sentiment"
] = preds_cross

final_predictions_df.to_csv(

    f"./{experiment_name}_predictions.csv",

    index=False
)

# =====
# SAVE METRICS
# =====

metrics_df = pd.DataFrame({

    "Metric": [

        "Accuracy",

        "Macro_F1",

        "Weighted_F1"
    ],

    "Score": [

        accuracy,

        macro_f1,

        weighted_f1
    ]
})

metrics_df.to_csv(

    f"./{experiment_name}_metrics.csv",
```

```

        index=False
    )

    # =====
    # CLEANUP
    # =====

    trainer.optimizer = None

    trainer.lr_scheduler = None

    del trainer

    del model

    gc.collect()

    torch.cuda.empty_cache()

    torch.cuda.ipc_collect()

    return {

        "accuracy": accuracy,

        "macro_f1": macro_f1,

        "weighted_f1": weighted_f1
    }

```

```

In [16]: metrics = run_experiment(

    "llama_epochs5",
)

```

```

[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Fetching 4 files: 100%|██████████| 4/4 [02:00<00:00, 30.09s/it]
Loading weights: 100%|██████████| 291/291 [00:08<00:00, 32.49it/s]

```

 [4690/4690 3:59:22, Epoch 5/5]

Step	Training Loss
20	1097.587988
40	1383.193066
60	246.941919
80	465.165479
100	194.130078
120	661.333936
140	81.904175
160	16.688290
180	31.046649
200	19.316656
220	55.725507
240	15.854295
260	29.199234
280	11.847794
300	24.253412
320	40.553430
340	18.020297
360	26.729428
380	33.882693
400	19.247009
420	13.126744
440	30.798178
460	214.385986
480	7.902782
500	11.544650
520	19.362537
540	9.559904
560	14.021799
580	6.912279
600	24.431889
620	25.989539
640	28.748016

Step	Training Loss
660	81.780939
680	9.710910
700	11.653049
720	79.381781
740	62.298615
760	18.206213
780	35.290723
800	8.223876
820	11.304639
840	13.539857
860	11.475824
880	13.541469
900	47.717465
920	15.248701
940	16.257117
960	12.562801
980	31.555145
1000	13.260088
1020	20.199237
1040	24.908870
1060	24.426999
1080	31.422620
1100	8.834453
1120	61.963428
1140	35.379065
1160	21.211237
1180	11.434863
1200	15.746629
1220	8.841106
1240	31.218457
1260	24.362645
1280	11.778678
1300	10.761227

Step	Training Loss
1320	17.488539
1340	50.082248
1360	45.915634
1380	12.991251
1400	8.664642
1420	7.953414
1440	14.211403
1460	13.557576
1480	14.129114
1500	29.388245
1520	13.957652
1540	17.949525
1560	9.090553
1580	13.521062
1600	66.482007
1620	8.463038
1640	31.834430
1660	51.186273
1680	12.302765
1700	16.464813
1720	7.085986
1740	9.553727
1760	36.009644
1780	10.402499
1800	6.066778
1820	48.410098
1840	4.665758
1860	22.630116
1880	9.929904
1900	32.795886
1920	36.454221
1940	33.401898
1960	27.666479

Step	Training Loss
1980	41.480856
2000	15.398157
2020	30.595746
2040	17.012642
2060	19.538042
2080	50.090161
2100	33.517700
2120	59.685254
2140	9.154534
2160	7.900117
2180	34.131369
2200	8.487579
2220	10.605068
2240	39.617468
2260	9.208611
2280	29.376120
2300	43.267160
2320	23.027544
2340	6.218105
2360	16.624594
2380	14.360260
2400	18.053790
2420	10.089272
2440	33.674606
2460	7.176905
2480	5.722142
2500	8.261637
2520	4.802241
2540	11.060354
2560	5.933010
2580	24.057457
2600	13.710342
2620	8.614807

Step	Training Loss
2640	22.086728
2660	5.052794
2680	5.593306
2700	23.768626
2720	26.362863
2740	55.322900
2760	8.754874
2780	9.901926
2800	9.460120
2820	17.192284
2840	17.187856
2860	4.901097
2880	7.806400
2900	53.058594
2920	13.713260
2940	15.488988
2960	22.562256
2980	13.478143
3000	165.536279
3020	26.477130
3040	19.641466
3060	5.911736
3080	8.219883
3100	23.656712
3120	13.487537
3140	24.753497
3160	43.089453
3180	59.091156
3200	83.509210
3220	58.226648
3240	10.375082
3260	7.741777
3280	21.998660

Step	Training Loss
3300	11.943986
3320	11.994816
3340	8.323733
3360	8.263181
3380	14.297412
3400	9.021815
3420	17.222981
3440	32.491473
3460	34.020190
3480	28.618991
3500	14.536223
3520	7.954465
3540	7.630055
3560	33.511597
3580	15.182191
3600	19.044484
3620	8.150874
3640	10.312119
3660	18.812646
3680	12.422786
3700	23.339050
3720	45.787643
3740	10.180263
3760	54.753253
3780	20.362032
3800	28.364407
3820	12.603046
3840	16.884325
3860	22.790800
3880	10.548212
3900	5.835807
3920	32.222275
3940	8.726147

Step	Training Loss
3960	4.991804
3980	8.455033
4000	4.866012
4020	17.985277
4040	4.657369
4060	5.176379
4080	22.361880
4100	26.431888
4120	8.111244
4140	10.106676
4160	9.876781
4180	9.949069
4200	48.840536
4220	12.086609
4240	4.175758
4260	10.590624
4280	8.244996
4300	7.369656
4320	10.578111
4340	21.647742
4360	339.984473
4380	22.614580
4400	14.819592
4420	13.848351
4440	6.136111
4460	20.688420
4480	17.678789
4500	7.572752
4520	17.361270
4540	11.657612
4560	6.869439
4580	20.111342
4600	143.429822

Step	Training Loss
4620	20.758810
4640	41.135913
4660	7.536687
4680	5.643608

```
[transformers] The following generation flags are not valid and may be ignored:
['temperature']. Set `TRANSFORMERS_VERBOSITY=info` for more details.
[transformers] Both `max_new_tokens` (=3) and `max_length` (=4096) seem to have be
en set. `max_new_tokens` will take precedence. Please refer to the documentation
for more information. (https://huggingface.co/docs/transformers/main/en/main\_classes/text\_generation)
```

```
===== CROSS DOMAIN RESULTS =====
```

[transformers] Ignoring clean_up_tokenization_spaces=True for BPE tokenizer TokenizersBackend. The clean_up_tokenization post-processing step is designed for Word Piece tokenizers and is destructive for BPE (it strips spaces before punctuation). Set clean_up_tokenization_spaces=False to suppress this warning, or set clean_up_tokenization_spaces_for_bpe_even_though_it_will_corrupt_output=True to force cleanup anyway.

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

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[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

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[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

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[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation

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for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

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[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

Accuracy: 0.8898104265402843

Macro F1: 0.6647491447020254

Weighted F1: 0.8770922450565026

Classification Report:

	precision	recall	f1-score	support
negative	0.95	0.54	0.69	132
neutral	0.46	0.31	0.37	36
positive	0.90	0.99	0.94	676
accuracy			0.89	844
macro avg	0.77	0.61	0.66	844
weighted avg	0.89	0.89	0.88	844

```
In [17]: metrics = run_experiment(
    "llama_lr3e-5",
    learning_rate=3e-5
)
```

Loading weights: 100%|██████████| 291/291 [00:08<00:00, 32.81it/s]

 [4690/4690 3:59:33, Epoch 5/5]

Step	Training Loss
20	1094.187988
40	1182.498926
60	109.840222
80	37.623529
100	9.091651
120	76.389703
140	11.497295
160	10.739279
180	27.184787
200	18.452925
220	75.022168
240	12.746278
260	27.917743
280	7.903599
300	38.131845
320	48.537265
340	18.967703
360	31.329654
380	29.484018
400	11.139972
420	13.431256
440	30.838483
460	181.928076
480	5.952268
500	11.167809
520	20.860905
540	8.771989
560	16.526508
580	6.740691
600	25.119427
620	23.904314
640	26.869208

Step	Training Loss
660	69.033398
680	7.450553
700	11.993906
720	68.836499
740	75.626343
760	17.849603
780	44.036420
800	7.254050
820	7.318303
840	11.643532
860	9.960890
880	13.671101
900	54.725739
920	14.437001
940	14.720070
960	13.240219
980	19.158755
1000	12.137505
1020	12.087409
1040	26.045517
1060	9.416401
1080	26.242130
1100	6.762913
1120	42.522076
1140	27.472690
1160	29.619727
1180	11.718811
1200	12.149578
1220	7.996770
1240	34.404053
1260	14.705579
1280	11.027952
1300	7.709253

Step	Training Loss
1320	11.847889
1340	67.698602
1360	36.316711
1380	12.928851
1400	9.584863
1420	7.538528
1440	13.698651
1460	11.368845
1480	14.395615
1500	30.923001
1520	10.081411
1540	11.905385
1560	7.546361
1580	10.408644
1600	65.095575
1620	6.471232
1640	31.050928
1660	52.842291
1680	11.837037
1700	9.786253
1720	7.103442
1740	8.711525
1760	26.231903
1780	10.835719
1800	4.884209
1820	39.885513
1840	3.797323
1860	23.303407
1880	10.882104
1900	33.929507
1920	35.411673
1940	29.447583
1960	30.474384

Step	Training Loss
1980	28.908167
2000	8.905086
2020	29.925955
2040	10.538988
2060	50.373837
2080	46.965530
2100	26.305078
2120	67.419232
2140	8.099556
2160	6.870653
2180	25.737790
2200	9.071082
2220	11.480549
2240	73.169763
2260	10.160726
2280	19.589160
2300	33.551450
2320	24.786021
2340	3.895945
2360	15.128645
2380	10.821317
2400	17.166972
2420	5.800196
2440	25.865939
2460	4.261129
2480	4.746821
2500	4.927938
2520	2.650342
2540	12.047136
2560	4.890135
2580	34.429483
2600	14.078047
2620	10.106876

Step	Training Loss
2640	24.154532
2660	3.532410
2680	5.038389
2700	15.496565
2720	33.414050
2740	30.074167
2760	4.061018
2780	9.624101
2800	15.296530
2820	16.484007
2840	23.622102
2860	3.019312
2880	2.843970
2900	39.997668
2920	15.974104
2940	4.151593
2960	21.145120
2980	10.370987
3000	174.805750
3020	17.849948
3040	6.465351
3060	4.893151
3080	4.119325
3100	35.637393
3120	8.366737
3140	11.808446
3160	15.017950
3180	66.325177
3200	97.849310
3220	56.175098
3240	2.870120
3260	5.538488
3280	25.961423

Step	Training Loss
3300	5.708746
3320	12.401016
3340	3.863718
3360	5.987144
3380	14.363857
3400	7.937421
3420	6.006753
3440	27.651328
3460	45.251883
3480	22.809445
3500	20.610481
3520	8.845457
3540	5.980120
3560	30.512811
3580	17.518469
3600	16.975421
3620	4.254082
3640	4.013586
3660	16.344350
3680	13.657883
3700	20.565819
3720	16.730183
3740	9.982931
3760	23.090192
3780	14.321454
3800	8.614290
3820	15.868300
3840	5.043540
3860	25.940436
3880	4.434791
3900	2.998178
3920	31.870630
3940	3.118328

Step	Training Loss
3960	3.680291
3980	9.061250
4000	1.734389
4020	3.870897
4040	3.865934
4060	3.251229
4080	10.062167
4100	18.234506
4120	5.490855
4140	10.813222
4160	8.360400
4180	10.138677
4200	99.765997
4220	11.575179
4240	3.128198
4260	3.198895
4280	6.607720
4300	5.204639
4320	8.799225
4340	15.649054
4360	19.273242
4380	18.124272
4400	15.098276
4420	13.955792
4440	2.058479
4460	12.421534
4480	21.401172
4500	3.733565
4520	12.926718
4540	9.523457
4560	9.396926
4580	8.979758
4600	156.753088

Step	Training Loss
4620	28.818564
4640	38.620169
4660	7.753896
4680	5.645134

[transformers] Both ``max_new_tokens`` (=3) and ``max_length`` (=4096) seem to have been set. ``max_new_tokens`` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

===== CROSS DOMAIN RESULTS =====

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[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

Accuracy: 0.8945497630331753

Macro F1: 0.6719181601803251

Weighted F1: 0.8802500842107953

Classification Report:

	precision	recall	f1-score	support
negative	0.94	0.55	0.70	132
neutral	0.59	0.28	0.38	36
positive	0.90	0.99	0.94	676
accuracy			0.89	844
macro avg	0.81	0.61	0.67	844
weighted avg	0.89	0.89	0.88	844

```
In [18]: metrics = run_experiment(
          "llama_3epochs",
          epochs=3
        )
```

Loading weights: 100%|██████████| 291/291 [00:08<00:00, 35.17it/s]

 [2814/2814 2:24:17, Epoch 3/3]

Step	Training Loss
20	1097.572168
40	1382.379102
60	247.223755
80	466.598975
100	196.805225
120	672.697021
140	84.827875
160	16.223167
180	31.007089
200	19.716812
220	52.540979
240	16.352766
260	28.354691
280	12.030101
300	23.524278
320	40.931992
340	17.861902
360	26.203259
380	34.255374
400	19.136487
420	12.816724
440	30.722952
460	212.117358
480	8.027636
500	11.412367
520	19.021167
540	9.386552
560	14.213341
580	6.965843
600	23.532460
620	26.012167
640	28.827646

Step	Training Loss
660	80.950684
680	9.970732
700	11.601655
720	77.034369
740	59.747272
760	18.396211
780	33.401038
800	8.140247
820	11.338916
840	13.325279
860	11.649261
880	13.496368
900	48.887827
920	15.242224
940	16.582275
960	12.775568
980	19.351816
1000	12.244684
1020	20.074728
1040	23.762144
1060	27.311313
1080	33.058527
1100	9.344702
1120	64.054474
1140	40.063678
1160	21.406003
1180	11.443472
1200	15.854895
1220	8.948489
1240	30.130045
1260	24.824323
1280	11.557513
1300	11.794372

Step	Training Loss
1320	18.087604
1340	50.701700
1360	44.592014
1380	12.661703
1400	8.347940
1420	8.086784
1440	14.615913
1460	13.741092
1480	14.539119
1500	32.512622
1520	13.766136
1540	17.837900
1560	9.062498
1580	15.504192
1600	61.718597
1620	8.477467
1640	33.409732
1660	49.556198
1680	11.748743
1700	16.846295
1720	7.209974
1740	9.707620
1760	33.006927
1780	10.604618
1800	6.617223
1820	45.120569
1840	4.809883
1860	23.892519
1880	10.018587
1900	33.876276
1920	37.547034
1940	31.293365
1960	25.258749

Step	Training Loss
1980	35.944254
2000	18.600032
2020	27.697678
2040	15.727025
2060	28.685107
2080	49.618936
2100	33.471616
2120	62.118011
2140	9.758706
2160	7.920583
2180	36.212103
2200	8.302397
2220	11.601401
2240	35.439514
2260	9.003081
2280	30.086707
2300	43.075058
2320	23.612791
2340	6.729870
2360	13.850311
2380	12.622469
2400	15.648988
2420	9.906786
2440	26.321832
2460	7.647802
2480	6.554556
2500	9.939865
2520	4.833163
2540	10.373203
2560	6.176595
2580	23.424754
2600	14.747252
2620	8.748653

Step	Training Loss
2640	21.222421
2660	5.536245
2680	6.084696
2700	27.245477
2720	23.201770
2740	67.819342
2760	9.508701
2780	9.681264
2800	8.485386

[transformers] Both `max\_new\_tokens` (=3) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

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[transformers] Both `max_new_tokens` (=3) and `max_length` (=4096) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main\_classes/text\_generation)
```

```
[transformers] Both `max_new_tokens` (=3) and `max_length` (=4096) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main\_classes/text\_generation)
```

```
[transformers] Both `max_new_tokens` (=3) and `max_length` (=4096) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main\_classes/text\_generation)
```

Accuracy: 0.8957345971563981

Macro F1: 0.6770972936470834

Weighted F1: 0.8814864544489719

Classification Report:

	precision	recall	f1-score	support
negative	0.95	0.54	0.69	132
neutral	0.58	0.31	0.40	36
positive	0.90	1.00	0.95	676
accuracy			0.90	844
macro avg	0.81	0.61	0.68	844
weighted avg	0.89	0.90	0.88	844

In []: